James Hampson

07342 327749 | Jameshampson04@outlook.com | LinkedIn | GitHub

PROFILE

Final-year Computer Science student at the University of Plymouth (expected First / high 2:1, May 2026) with hands-on experience across the full stack — from custom OpenGL rendering pipelines to mobile GPS engines and real-time Firebase backends. Delivered production-ready software to a real client in a 4-person team.

PROJECTS

Mobile Racing App | Java · Android Studio · Firebase · GPS **[Oct 2024 – Jun 2025]**

- Delivered a GPS racing game to a real client as part of a 4-person Agile team — coordinating across location, backend, and rendering responsibilities from spec to hand-off
- Engineered the core location system using Android GPS APIs, linking real-time position updates to scoring and game progression
- Designed and built a custom map tile renderer from open-source geodata, removing third-party map dependencies, which allowed offline play and avoided licensing costs
- Integrated Firebase Auth and Realtime Database to support live matchmaking, user accounts, and leaderboards across concurrent game sessions

3D Graphics Engine | C++ · OpenGL · GLSL **[Jan 2026 – May 2026]**

- Implemented a complete real-time 3D rendering pipeline in C++ and OpenGL, covering geometry processing, vertex transformation, and fragment output
- Authored custom GLSL vertex and fragment shaders with Phong/Blinn-Phong lighting, modelling diffuse, specular, and ambient components independently to achieve physically grounded shading
- Constructed a multi-object scene with OBJ model loading and UV texture mapping, managing per-object material state and draw calls within a single render loop
- Built a first-person camera system using matrix-based view and projection transforms, handling real-time input to update orientation and position each frame

XR Interaction Project | C# · Unity · XR Toolkit **[Nov 2025 – Present]**

- Architected a modular grid-based puzzle system in Unity, decoupling level layout from interaction logic to allow rapid scene iteration without rewriting game rules
- Implemented 6DOF controller-driven object interaction using the XR Toolkit, handling grab, release, and placement events with real-time haptic and visual feedback

EDUCATION

BSc (Hons) Computer Science - University of Plymouth, expected high 2:1 to First **[2023 – 2026]**
Digital Production, Design & Development - Strode College, Merit **[2021 – 2023]**

SKILLS

Languages: C#, C++, Java, Python
Frameworks & Tools: Git, Unity, Android SDK, Firebase, OpenGL, Docker, SQL/databases, RESTful APIs, GLSL
Concepts: Object-Oriented Programming, Data Structures & Algorithms, Agile/Scrum Development

WORK EXPERIENCE

Store Assistant — Home Bargains (Part-time, 19 months)
IT Assistant — Somerset Council Library (Voluntary, 14 months)